

CHEMISTRY XI

10 Year Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF CHEMISTRY.

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QUESTIONS	YEARS
Chapter 1: STOICHIOMETRY	
Section B	
ii) Calculate the volume of CO_2 at $25^\circ C$ and 780 torr pressure when 12.45 grams of marble is heated: $CaCO_3 \rightarrow CaO + CO_2$ —OR— 200 grams of $K_2Cr_2O_7$ reacted with 200 grams of concentrated H_2SO_4 . Calculate: *Mass of atomic oxygen produced *Mass of reactant left unreacted $K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3[O]$	2025
ii) Chlorine gas is produced on an industrial scale by electrolysis of a aqueous solution of $NaCl$ $2NaCl + 2H_2O \rightarrow Cl_2 + H_2 + 2NaOH$ Calculate; *The volume of Cl_2 at S.T.P produced from 58.5 g of $NaCl$. *The total number of molecules of Cl_2 produced. iii) 53.5g of Aluminium is burnt with 22.4g of oxygen to given brilliant white flame of Al_2O_3 . $4Al + 3O_2 \rightarrow 2Al_2O_3$ Which reactant is limiting reactant? Also find the mass of Al_2O_3 produced.	2024
2. ii) $AlCl_3$ is produced by the following reactant at $500^\circ C$ $2Al + 3Cl_2 \rightarrow 2AlCl_3$ When 160gm of Aluminium is reacted with excess of chlorine gas, 650gm of $AlCl_3$ is obtained. Calculate (a) Theoretical yield (b) % yield 2. iii) Combustion of Ethene (C_2H_4) in air to form CO_2 & H_2O	2023

is given in the following equation: $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$ If a mixture containing 2.8 gm C_2H_4 & 6.4 gm O_2 is allowed to ignite, identify the limiting reactant and determine the mass of CO_2 gas that will be formed.	
(ii) Calculate the volume of CO_2 gas produced at $25^\circ C$ and 740 torr pressure by the combustion of 20 grams of CH_4 $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	2022
2. iv) If 53.5g of NH_4Cl is heated with $Ca(OH)_2$, how many grams of NH_3 is produced? Also find the volume of NH_3 at S.T.P according to the following equation. $2NH_4Cl + Ca(OH)_2 \longrightarrow 2NH_3 + CaCl_2 + 2H_2O$	2021
(ii) Zinc reacts with H_2SO_4 (dil) as given below: $Zn + H_2SO_4 \xrightarrow{dil} ZnSO_4 + H_2$ Calculate the mass of $ZnSO_4$, the volume of H_2 gas at S.T.P. and the number of molecules of H_2 gas which will be produced by reacting 163.5g of Zn with H_2SO_4.	2019
(ii) Find the mass of CH_3OH produced. When 356 gm of CO is mixed with 65.0 gm of H_2 . $CO + 2H_2 \longrightarrow CH_3OH$	2018
(iii) (a) Simplify the following by using rules of significant figures: $\frac{92 \times 751 \times 173}{760 \times 297}$ (iv)(a) 10 gm of H_2SO_4 has been dissolved in excess of water to dissociate it completely into ions. Calculate the : * Number of molecules in 10 gm of H_2SO_4 * Number of Positive Ions (iv) (b) Define the following: * Limiting reactant	2016
(ii) The empirical formula of compound is CO_2H . 1.8 g of this compound in gaseous state occupies 448 cm^3 at S.T.P find its molecular formula.	2015
(ii) $ZnCl_2$ is prepared by the reaction $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ 6.54 gram of Zn reacts with 73 gram of HCl. Find the limiting reactant and the mass of $ZnCl_2$ produced. (iii) (a) The mass of a substance is 18.8865 grams and its volume is 7.9 cm^3 . what will be its density considering significance figure and rounding off the above obtained? (vi) Calculate the volume of Nitrogen gas produced by heating 800 grams of Ammonia at $21^\circ C$ and 823 torr pressure: $2NH_3 \rightleftharpoons N_2 + 3H_2$	2014

Section C

4. (c) Combustion of 0.5 gm of a Hydrocarbon produced 1.515 gm CO_2 and 0.77 gm of H_2O . If the molecular mass of the compound is 58 a.m.u., determine its Molecular formula. **2014**

Chapter 2: ATOMIC STRUCTURE

Section B

- iii) State Pauli's Exclusion principle. Write the electronic configuration of the following: **2025**
 $^*\text{Ca}^{++}(\text{Z} = 20)$ $^*\text{Br}^-(\text{Z} = 35)$ $^*\text{Ag}(\text{Z} = 47)$
- iv) State 'Hund's rule'. Write electronic configuration of the following: **2024**
 $^*\text{Cr}(\text{Z} = 24)$ $^*\text{Sr}^{++}(\text{Z} = 38)$ $^*\text{S}^{--}(\text{Z} = 16)$
2. iv) State Pauli's Exclusion Principle. Write electronic configuration of the following: **2023**
 $^*\text{Cl}^-(\text{Z} = 17)$ $^*\text{Ca}^{+2}(\text{Z} = 20)$ $^*\text{Cu}(\text{Z} = 29)$
- (vi) State $(n + l)$ rule and write the electronic configuration of the following: **2022**
 $^*\text{Mo}(\text{Z} = 42)$ $^*\text{S}(\text{Z} = 16)$ $^*\text{Sr}^{++}(\text{Z} = 38)$
2. vi) State Pauli's Exclusion Principle. Write the electronic configuration of the following: **2021**
 $^*\text{Ca}^{++}(\text{Z} = 20)$ $^*\text{Cr}(\text{Z} = 24)$ $^*\text{Br}(\text{Z} = 35)$
- (vii) State $(n + 1)$ rule. Write the electronic configuration of the following: **2019**
 $^*\text{Cu}(\text{Z} = 29)$ $^*\text{Sr}^{++}(\text{Z} = 38)$ $^*\text{O}^{--}(\text{Z} = 16)$
- (vii) State Pauli's exclusion principle and write the electronic configurations of the following: **2018**
 $^*\text{Ca}^{+2}(\text{Z} = 20)$ $^*\text{Br}^-(\text{Z} = 35)$ $^*\text{Cr}(\text{Z} = 24)$
- (vii) What is radioactivity? Why is it limited to heavy elements? Write the properties of Alpha particles. **2017**
- (xii)(a) Write the electronic configuration of the following: **2016**
 $^*\text{Cu}^+(\text{Z} = 29)$ $^*\text{Br}^-(\text{Z} = 35)$
 (b) Using $n + l$ rule, arrange 4d, 7s and 4f energy levels in ascending order of energy
- (xii)(a) Write the electronic configuration of the following: **2015**
 $^*\text{Cu}(\text{Z} = 29)$ $^*\text{Ag}(\text{Z} = 47)$
 (b) Which rule of principle is violated in the following electronic configurations?
 $^*1s^2, 2s^3$ $^*2s^2, p^6, 3s^2$ $^*1s^2, 2s^2, 3s^2, 2p^6$ $^*1s^2, 2s^2, 2p^2$
- (v)a. Write the electronic configuration of the following **2014**

* Cr(Z=24) * Mo (Z=42)

(viii) Explain Rutherford's Atomic model with its conclusion.

Section C

3. a) what are X-rays? How are they produced? Give any four properties of X-rays also derive an expression for the energy of electrons in n^{th} orbit according to Bohr's Model of atom when: $r = \frac{\epsilon_0 n^2 h^2}{Z \pi m e^2}$	2025
3. a) Write the postulates of Bohr's atomic model. Starting from $\Delta E = E_2 - E_1$ derive an expression for the wave number of radiations emitted by an electron.	2024
3. a) Write the four main postulates of Bohr's atomic theory. Derive the relation $r_n = \frac{a^0 n^2}{Z}$ for the radius of n^{th} orbit of Hydrogen atom.	2023
3. a) (OR) Write the postulates of Bohr Atomic Theory. Derive an expression for radius of n^{th} orbit in hydrogen atom. 5. b) (OR) What is Natural Radioactivity? Write the properties of α and β particles	2021
3. (a) Starting from $\Delta E = E_2 - E_1$ derive the expression for the wave number for hydrogen atom. 3. (b) Define Radioactivity. Describe the character of Alpha and beta rays.	2019
3. (b) State the postulates of Bohr's Atomic Theory and derive an expression for the energy of an electron in the n^{th} orbit when $r = \frac{n^2 h^2}{4 \pi^2 m Z e^2}$	2018
3. (a) State postulates of Bohr's atomic model. Derive expression for the frequency when the electron jumps from higher orbit (n_2) to lower orbit (n_1). The energy of the electron $E = \frac{2 \pi^2 Z^2 e^4 m}{n^2 h^2}$ 5. (b) State Hund's rule and write the electronic configurations of the following: *Cl ¹ (Z=17) *Cr (Z=24) *Rb ¹ (Z=37)	2017
3. (b) Derive an expression for the radius of n^{th} orbit for Hydrogen atom using Bohr's atomic model. (c) Explain the origin of X- Rays and describe the relationship between their wavelength and atomic number.	2016
3(b) Give the defects of Rutherford's Atomic Theory. Write the postulate of Bohr's Atomic Theory 3(c) How did Bohr's theory explain the formation of the line spectrum of hydrogen atom? Write the names and formulae of	2015

each series.	
5.(a) Starting from $\Delta E = E_2 - E_1$ drive the expression for the wave number for the hydrogen atom.	2014

Chapter 3: THEORIES OF COVALENT BONDING AND SHAPES OF MOLECULES

Section B

xiv) Draw molecular orbital diagram of O_2 molecules. Write molecules orbital configuration and determine its bond order.	2025
2. v) Draw the shape of BF_3 , NH_3 , CH_4 and H_2O molecules on the basis of VSEPR theory.	2023
(x) Define and explain orbital hybridization. Describe sp^2 hybridization in C_2H_4 .	2022
(viii) What is Covalent bond? Explain the ionic character of Covalent bond.	2019
(viii) Explain ionic character of Covalent bond.	2018
(iii) (b) Draw the electronic structures of ; * Phosphorus oxychloride * Nitromethane	2016
(ix) State the main postulates of Electron pair repulsion theory & explain shape of NH_3 according to this theory.	2015
(b) Differentiate between Sigma and Pi bonds (xi) What is electro negativity? Explain the ionic character of covalent bond with its help	2014

Section C

4. a) Discuss the shapes of the following molecules with the help of hybridization and VSEPR. * BF_3 * $BeCl_2$ * NH_3	2025
4. a) Write the postulates of VSEPR model. State the shapes of the following molecules with the help of hybridization * H_2O * BF_3	2024
3. b) What is Hybridization? Write its types. CH_4 , NH_3 & H_2O all have same hybridization but their molecular structures are different. Explaining their structures give a reason for this difference.	2023
5. (a) Explain the structures of BF_3 and NH_3 on the basis of electron pair repulsion model and hybrid orbital model.	2022
4. a) Write the postulates of VSEPR Theory. Explain the shape of the molecules which have: * three bond pair electron * two bond pair and two lone pair	2021

4. (a) Define Orbital Hybridization. Explain the shape of Ethene, C_2H_4 on the basis of Hybridization.	2019
4. (b) Explain the structure of $BeCl_2$ and NH_3 on the basis of Electron pair repulsion model and Hybrid orbital model.	
4. (b) Give the postulates of electron pair repulsion model and draw the molecular shape of BF_3 and H_2O on the basis of this model	2018
3. (b) Define Hybridization and write its types. Explain hybridization in Ethane, C_2H_6 or Ethene, C_2H_4 molecules.	2017
5. (a) Draw and explain the geometrical shapes of molecules with the following number of electrons pair: * Two bond pairs * Three bond pairs and one lone pair * Two bond pairs and two lone pairs.	
4. (b) Explain the shapes of the following molecules on the basis of Electron pair repulsion model and hybrid orbital model: (i) H_2O (ii) $BeCl_2$	2016
3. (a) Define Orbital Hybridization. Explain Sp^2 hybridization with example. 3. (b) Distinguish between: * Atomic orbital and Molecular orbital	2015
4. (a) Explain the shapes of BF_3 and H_2O on the basis of: Hybrid orbital model and Electron pair repulsion model.	2014

Chapter 4: STATE OF MATTER: GASES

Section B

v) What is ideal gas? Drive the general gas equation. v) –OR– Write the postulates of Kinetic Molecular Theory of gases. xiii) If 255cm^3 oxygen gas is collected over water at 25°C and a total pressure of 785 torr. Then find the mass of dry oxygen gas (Pressure of water vapour at 25°C is 24 torr) vi) One mole of NH_3 gas occupies the volume of one dm^3 at 25°C , assuming NH_3 as a real gas, determine its pressure, while: $a = 4.170\text{ dm}^6\text{atm mol}^{-2}$ $b = 0.0371\text{ dm}^3/\text{mol}$	2025
vi) A 100cm^3 sample of hydrogen effuse four times as rapidly as 100cm^3 of an unknown gas. Calculate the molecular mass of the unknown gas. vii) Drive Van der Waal's equation for real gases by correcting volume and pressure.	2024
2. vii) 40 dm^3 of hydrogen gas is collected over water at 23°C	2023

and 831 torr pressure. What would be the volume of dry gas obtained at STP (vapour pressure of water at 23°C = 21 torr.	
2. ix) State Graham's Law of diffusion. The ratio of rates of diffusion of two gases A and B is 1.5:1 respectively. If the relative molecular mass of gas A is 16, find the molecular mass of B gas.	
2. xiv) (OR) What is the fourth state of matter? State about its occurrence, characteristics and uses.	
(iv) What is Ideal Gas, Derive the General gas equation.	2022
(v) Calculate the density of SO ₂ in g/dm ³ at 25°C and 300 torr pressure.	
2. ii) Write the postulates of Kinetic Molecular Theory of gases.	2021
2. x) (OR) 1.40 dm ³ volume of gas collected at a temperature of 27°C and a pressure of 900 torr was found to have a mass of 2.273g. Calculate molecular mass of the gas.	
(iv) A 500 cm ³ vessel contains H ₂ gas at 400 torr pressure and another 1 dm ³ contains O ₂ gas at 600 torr pressure. If these gases are transferred to 2 dm ³ empty vessel, calculate the pressure of the mixture of gases.	2019
(v) What is Ideal gas? Derive the ideal equation.	
(iii) 100 gram of KNO ₃ heated to redness. Find volume of oxygen is liberated at 39° C and 765 torr pressure.	2018
$2\text{KNO}_3 \longrightarrow 2\text{KNO}_2 + \text{O}_2$	
(v) Calculate the density of SO ₂ in g/dm ³ at 25°C and 300 torr pressure.	
(vi) What is Ideal gas? What is Ideal gas? Derive the ideal gas equation.	
(iii) $\text{CaCO}_3 + \text{HCl} \xrightarrow{\text{Heat}} \text{CaO} + \text{CO}_2$	2017
CaCO ₃ is often used to generate CO ₂ gas in industry, if 200g of CaCO ₃ , is strongly heated, what volume of CO ₂ gas will be obtained at 30°C and 1200 torr pressure?	
(v) A gaseous mixture of H ₂ and O ₂ at 60°C and total pressure of 2.5 atmospheres contains 75 mole H ₂ and 25 moles of O ₂ . What are the partial pressures of both the gases in the mixture?	
(vi) State the relevant gas laws which explain the following observations of gases:	
* When a balloon filled with air is kept in cold place at constant pressure, its volume reduced.	
* When the pressure on a gas kept in an airtight cylinder at	

constant temperature is doubled, its volume becomes half.	
(ii) What volume of CO_2 measured at 20°C and 720 torr pressure will be produced by the reaction between 200 gm of Na_2CO_3 and HCl $\text{Na}_2\text{CO}_3 + \text{HCl} \longrightarrow 2\text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$	2016
(vi) 1100 ml of CO_2 at a pressure of 500 torr, 1500 ml N_2 at a pressure of 400 torr and 800 ml of O_2 at a pressure of 600 torr are placed together in a container of 1000 ml capacity. Find the partial pressure of each gas.	
(vii) H_2 gas effuses from a 1dm^3 vessel in 10 sec. Calculate the time for effusion of O_2 gas from 20dm^3 vessel.	
(ix) (a) If 16 ml of Hydrogen diffuse in 30 sec, what volume of SO_2 will diffuse in the same time (30 Sec.) Under similar conditions?	
(iv) A gaseous mixture contains 0.2 mole of O_2 and 0.3 mole of CO_2 . If the partial pressure of oxygen is 60 torr, calculate the partial pressure of CO_2	2015
(vi) Calculate the volume of Nitrogen gas produced by heating 800 grams of Ammonia at 21°C and 823 torr pressure:	2014
(xi) What is an ideal gas? What are the causes of deviation of the real gases from ideal behavior?	
(xv) State and explain Dalton's Law of Partial pressure with its applications.	
Section C	
4. (c) State Boyle's, Charles's and Avogadro's Laws and derive an expression for the equation of state. Calculate the value of Gas constant in two different units	2016
3(c) Write the postulates of kinetic molecular theory of gases.	2015
5. (c) The density of a certain gas is 1.43 gram/dm^3 at 608 and 27°C . Find the molecular mass of the gas.	2014

Chapter 5: STATE OF MATTER: LIQUIDS

Section B

Section C

5. a) Explain the properties of water, based on hydrogen bonding? Write the properties of crystalline solid.	2025
5. b) ---OR--- Write note on any one of the following: *Viscosity *Surface Tension *London Dispersion Forces	

Chapter 6: STATE OF MATTER: SOLIDS

Section B

Section C

5. a) Explain the properties of water, based on hydrogen bonding? Write the properties of crystalline solid.	2025
5. b) Give the characteristics of ionic and covalent solids. K_c for reaction is 55 at 700°C . Calculate the equilibrium concentration of reactants and products when the concentration of each reactant is 2.55 mole/dm^3 . $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$	2024
3. (c) What is Unit cell? Sketch a unit cell and label the dimensions a, b, c and angle α, β, γ . How do cubic and tetragonal system differ?	2019
3. (c) Differentiate between Isomorphism and Polymorphism and explain Ionic crystal.	2018
4. (b) What is Unit cell? Sketch a unit cell and label the dimension a, b, c, and angles α, β, γ . How do cubic and tetragonal system differ?	2017

Chapter 7: CHEMICAL EQUILIBRIUM

Section B

vii) State Le-Chatelier's Principle in Haber's process: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3 (\Delta H = -ve)$	2025
viii) The solubility of $\text{Mg}(\text{OH})_2$ at 25°C is $4.6 \times 10^{-3} \text{ g/100cm}^3$, calculate the solubility product of the $\text{Mg}(\text{OH})_2$.	2024
2. x) Define Solubility Product. Write the solubility product expression for the following: * PbCl_2 * $\text{Fe}(\text{OH})_3$ * Ag_2SO_4	2023
(viii) Define Solubility. The solubility of $\text{Mg}(\text{OH})_2$ at 25°C is 0.0046 gm/dm^3 . What is the solubility product of $\text{Mg}(\text{OH})_2$? (ix) State and explain Le-Chatelier principle with reference to, the formation of NH_3 by Haber's process or sulphuric acid by contact process.	2022
2. viii) 9.2 of ethyl alcohol and 12g of acetic acid kept at constant temperature until equilibrium was established 4.0g of acid was left unused. Calculate K_c for the reaction: $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$	2021
(xi) State le-Chatelier's Principle. Give its applications in the	2019

manufacture of SO_3 by Contact process.	
(xii) The K_c for the reaction $2HI \rightleftharpoons H_2 + I_2$ is 1.3×10^{-2} . If there are 0.5 moles/dm^3 of H_2 , 1.5 moles/dm^3 of I_2 and 5 moles/dm^3 HI , predict the direction in which the reaction moves so as to achieve the equilibrium.	
(xi) State and explain the law of mass action. Derive the expression for the general reversible reaction given below: $mA + nB \rightleftharpoons xC + yD$	2018
(xii) 4.6 gram of ethyl alcohol and 6.0 gram of acetic acid were kept at constant temperature until equilibrium was established. 2.0 gram of acetic acid remained unused. Calculate K_c for the reaction. $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$	
(x) State Le-Chatelier's Principle. Give an application on the manufacture of SO_3 by Contact process.	2017
(xi) Phosphorus pentachloride decomposes in a gas phase reaction at 250°C as follows: $2PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ An equilibrium mixture in a 5.0 liter container is found to have 3.84 gm PCl_5 , 9.14 gm PCl_3 and 2.84 gm of Cl_2 . Evaluate K_c at 250°C .	
(vii) Define solubility. The solubility of $Mg(OH)_2$ at 25°C is $4.6 \times 10^{-3} \text{ gm/100 cm}^3$. What is the solubility product of $Mg(OH)_2$?	2016
(viii) For the reaction: $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ the equilibrium mixture contains 0.25 M Nitrogen and 0.15 M Hydrogen gas at 25°C . Calculate the concentration of NH_3 . Given $K_c = 9.6$. The volume of the container is 1 dm^3 .	
(xv) (a) Give the relationship between K_p and K_c .	
(viii) In the reaction $H_2 + I_2 \rightleftharpoons 2HI$, When equilibrium was attained, the concentrations were $[H_2] = [I_2] = [HI] = 4 \text{ moles/dm}^3$. Calculate the equilibrium constant and the initial concentration for H_2 and I_2 .	2015
(vi) What are the ionic concentrations of Ag^+ and CrO_4^{2-} in a saturated solution of Ag_2CrO_4 at 25°C ? K_{sp} of Ag_2CrO_4 is $1.9 \times 10^{-12} \text{ mole}^3/\text{dm}^9$.	
(x) 5.88 moles of Nitrogen and 16.2 moles of Oxygen are mixed and heated at 2000°C until the equilibrium is established, 11.28 moles of Nitric oxide are formed. Calculate the value of the equilibrium constant.	2014

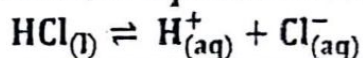
$\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$ (xii) State the law of Equilibrium. Derive the expression of K_c for the following reaction: $m\text{A} + n\text{B} \rightleftharpoons x\text{C} + y\text{D}$	
Section C	
3. b) State Law of Mass Action. Derive an expression of equilibrium constant K_c for general reversible reaction: $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$ Find equilibrium constant when 2.94 moles of Iodine and 8.1 moles of hydrogen were mixed and heated at 444°C until the equilibrium was established 5.64 moles of HI were formed: $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$	2025
4. b) State law of mass action. Derive the expression of K_c for the general reversible reaction $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$ and also give the relationship between K_p, K_c and K_A. 5. b) Give the characteristics of ionic and covalent solids. K_c for reaction is 55 at 700°C. Calculate the equilibrium concentration of reactants and products when the concentration of each reactant is 2.55 mole/dm^3. $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$	2024
3. (b) 1 mole of N_2 and 1 mole of O_2 were heated at 2000°C in 10 dm^3 vessel. Calculate the equilibrium concentration of Nitric Oxide K_c for the reaction is 0.1 $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$	2022
5. b) State the law of mass action. Derive K_c for general reversible reaction and predict the extent of reaction.	2021
5. (c) Will Cadmium hydroxide precipitate from 0.02 M solution of CdCl_2 at $\text{pH}=10$? $K_{sp} \text{ of } \text{Cd}(\text{OH})_2 = 2.5 \times 10^{-14} \text{ mole}^2 / \text{dm}^6$	2019
4. (c) Should AgCl precipitate form a solution prepared by mixing 400 ml of 0.1 M NaCl and 600 ml of 0.03M AgNO_3? (K_{sp} of $\text{AgCl} = 1.6 \times 10^{-10} \text{ mole}^2 / \text{dm}^6$)	2018
3. (c) The concentration of Ag^+ ions is $1 \times 10^{-6} \text{ M}$ in AgBr solution. Find the minimum concentration of Br^- ions necessary to cause precipitation of AgBr (K_{sp} of $\text{AgBr} = 4 \times 10^{-13} \text{ M}^2$)	2017
5 (c) State and explain the law of Mass action. Derive the expression of K_c for the general reversible reaction.	2016
5.(a) State Le-Chatelier's principle. Apply this principle to the manufacture of NH_3 by Haber's Process.	2015
(v)b. Calculate the solubility product of PbCrO_4 when the	2014

solubility of PbCrO_4 is 1.0×10^{-3} grams/ dm^3 .**Chapter 8: ACID BASE AND SALTS****Section B**

viii) Explain Lewis Theory of acids and bases with example. 2025

xii) Explain Buffer solution and write its applications.

ix) Explain Bronsted Lowry Theory of acids and bases. Also explain conjugate acid-base pair. 2024

2. xi) Define pH and pOH of a solution and write its formula. Calculate pOH of 0.0025M aqueous solution of HCl at 25°C . 2023(xiii) Define pH . The pH of solution is 8.4. Find: 2022* pOH of the solution * Hydrogen ion concentration

* Hydroxyl ion concentration

(xii) Define pH and pOH . Calculate the pH and pOH of a solution of Nitric acid whose concentration is 2.0×10^{-2} mole/ dm^3 2017(xiv) The aqueous solution of 0.3M $\text{Mg}(\text{OH})_2$ is 60% ionized. Calculate its pH . 2016(xi) Define pH . What is the pH of 0.002M NaOH solution at 25°C ? 2015**Section C****Chapter 9: CHEMICAL KINETICS****Section B**

xi) Explain the effect of the following factors on rate of a chemical reaction. 2025

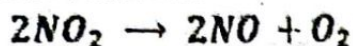
* Concentration of reactant * Temperature

x) The reaction $2\text{NO} + \text{Cl}_2 \rightarrow 2\text{NOCl}$ was studied at 25°C the following results were obtained: 2024

Exp. No.	Initial Concentration mol/ dm^3		Initial rate mol/ dm^3sec
	[NO]	[Cl ₂]	
1	0.1	0.1	2.52×10^{-3}
2	0.1	0.2	5.04×10^{-3}
3	0.2	0.1	10.05×10^{-3}

On the basis of above information illustrate the Rate law and find the order of reaction.

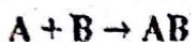
2. xiii) What is order of reaction? Decomposition of NO_2 into NO and O_2 is 2nd order reaction:



If the rate constant at certain temperature is $3.8 \times 10^{-4} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and initial concentration of NO_2 is 0.38M, calculate the initial rate of reaction.

2023

(xii) Write the rate law. Determine the order of reaction using the following data:



Exp.	[A]	[B]	$\frac{dx}{dt}$
1	0.1M	0.1M	$2 \times 10^3 \text{ MS}^{-1}$
2	0.2M	0.1M	$8 \times 10^3 \text{ MS}^{-1}$
3	0.1M	0.2M	$4 \times 10^3 \text{ MS}^{-1}$

2022

2. x) Discuss the effect of following factors on rate of reactions.

* Concentration

* Radiation

2021

(xiv) Define Activation energy. Give its relationship with the speed of reactions.

(xv) Determine the order of reaction:

$\text{A} + \text{B} \longrightarrow \text{Product}$ From the following data:

S.No.	A moles/ dm^3	B moles/ dm^3	Rate MS^{-1}
1	0.1	0.1	1×10^{-3}
2	0.2	0.1	4×10^{-3}
3	0.1	0.3	3×10^{-3}

2019

(xiv) Determine the order of reaction from the data given below:



Experiment Number	[A] mole/ dm^3	[B] mole/ dm^3	Rate of reaction mole/ dm^3
1	0.1	0.1	3.0×10^{-3}
2	0.2	0.1	6.0×10^{-3}
3	0.1	0.3	9.0×10^{-3}

2018

(xv) Make the list of factors which affect the rate of reaction.

Discuss the following factors:

*Surface area

*Temperature:

(xiii) The rate constant k for the decomposition of Nitrogen dioxide is $1.8 \times 10^{-3} \text{ dm}^3 \text{ mole}^{-1} \text{ sec}^{-1}$ $2\text{NO}_2 \longrightarrow 2\text{NO} + \text{O}_2$

* Write the rate expression.

*Find the initial rate, when the initial concentration of NO_2 is 0.75M.

* Find the rate constant k when the initial concentration of NO_2 is doubled.

2017

(b) For the reaction : $2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$, calculate the rate constant when the initial concentration of N_2O_5 is 10.8g/dm^3 while the initial rate is $2.96 \times 10^{-4} \text{ mole/dm}^3 \cdot \text{Sec}$.
(xiv) How are the nature and surface area of the reactant related to the rate of reaction

2016

(xiii) (a) Define Activation energy. Give its relation with the speed of reaction.

2014

(xiii) (b) For the reaction $A \longrightarrow \text{Product}$, threshold energy is 40 KJ/mole . The average internal energy of the reactants is 10 KJ/Mole . Calculate its activation energy.

Section C

5. b) Describe the types of colloids. Write their properties. A certain reaction $A + B \rightarrow \text{Product}$ gave the following results:

2025

Exp. No	Initial Cone. mole/dm^3		Initial rate $\text{mole/dm}^3 \cdot \text{s}$
	[A]	[B]	
1	1.0×10^{-6}	9.0×10^{-6}	1.98×10^{-4}
2	2.0×10^{-6}	9.0×10^{-6}	3.96×10^{-4}
3	1.0×10^{-6}	3.0×10^{-6}	6.6×10^{-5}

On the basis of above results find the Rate Law and the order of reaction.

3(c) For the chemical reaction $\text{F}_2 + 2\text{ClO}_2 \longrightarrow 2\text{FCIO}_2$

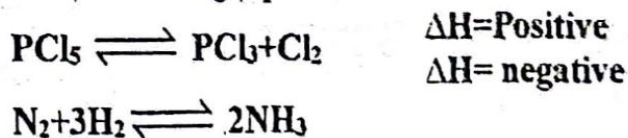
2015

Calculate the:

(i) Rate expression (ii) Order of Reaction (iii) Rate constant when the initial concentration of F_2 is 0.1 mole/dm^3 , ClO_2 is 0.01 mole/dm^3 and rate of reaction is $1.2 \times 10^{-3} \text{ mole/dm}^3 \cdot \text{sec}$

3. (c) Predict the effects of change in temperature and pressure on the following equilibrium:

2014



Chapter 10: SOLUTION & COLLOIDS

Section B

ix) State Raoult's Law and drive its mathematical expression.

2025

x) A glucose solution contains 0.15 mole of Glucose and 5.8 moles water at 20°C . Determine the lowering in the vapour

pressure if vapour pressure of pure solvent at 20°C is 17.5 torr(assume solution is an ideal)																			
xi) Define Colligative properties. Explain elevation of boiling point and depression of freezing point. xiv) At 30°C the vapour pressure of water is 31.6 torr. If 30g of urea (molecular weight = 60) are dissolved in 180g of water. Calculate the lowering of vapour pressure.	2024																		
2. xiv) State Raoult's Law. Derive this law mathematically in three forms. How this law can be applied on binary solution of miscible liquids?	2023																		
(iii) (b) Calculate the molarity of solution prepared by dissolving 4.5 gm of NaOH in 500cm ³ of water.	2014																		
Section C																			
5. b) Describe the types of colloids. Write their properties. A certain reaction $A + B \rightarrow \text{Product}$ gave the following results:	2025																		
<table><tr><th rowspan="2">Exp. No</th><th colspan="2">Initial Cone. <i>mole/dm³</i></th><th rowspan="2">Initial rate <i>mole/dm³.s</i></th></tr><tr><th>[A]</th><th>[B]</th></tr><tr><td>1</td><td>1.0×10^{-6}</td><td>9.0×10^{-6}</td><td>1.98×10^{-4}</td></tr><tr><td>2</td><td>2.0×10^{-6}</td><td>9.0×10^{-6}</td><td>3.96×10^{-4}</td></tr><tr><td>3</td><td>1.0×10^{-6}</td><td>3.0×10^{-6}</td><td>6.6×10^{-5}</td></tr></table>	Exp. No	Initial Cone. <i>mole/dm³</i>		Initial rate <i>mole/dm³.s</i>	[A]	[B]	1	1.0×10^{-6}	9.0×10^{-6}	1.98×10^{-4}	2	2.0×10^{-6}	9.0×10^{-6}	3.96×10^{-4}	3	1.0×10^{-6}	3.0×10^{-6}	6.6×10^{-5}	
Exp. No		Initial Cone. <i>mole/dm³</i>			Initial rate <i>mole/dm³.s</i>														
	[A]	[B]																	
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3	1.0×10^{-6}	3.0×10^{-6}	6.6×10^{-5}																
On the basis of above results find the Rate Law and the order of reaction.																			
5. (c) 4.0 g NaOH is dissolved in 2.5 dm ³ solution. Find its molarity and pH.	2018																		
5. (c) 19.6 gm of H ₂ SO ₄ is dissolved in 50cm ³ water to make a solution. What mass of NaOH should be dissolved in 250 cm ³ of water to prepare the solution of the same molarity as that of H ₂ SO ₄ .	2017																		

Chapter 11: THERMOCHEMISTRY

Section B

xii) Calculate the Heat of formation of Fe ₂ O ₃ at 25°C: $* 2\text{Fe} + \frac{3}{2}\text{O}_2 \rightarrow \text{Fe}_2\text{O}_3 \quad \Delta H_f^\circ = ?$ $* \text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O} \quad \Delta H = -286 \text{ KJ/mole}$ $* \text{Fe}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow 2\text{Fe}(\text{OH})_3 \quad \Delta H = +289 \text{ KJ/mole}$ $* \text{Fe} + 3\text{H}_2\text{O} \rightarrow \text{Fe}(\text{OH})_2 + \frac{3}{2}\text{H}_2 \quad \Delta H = +161 \text{ KJ/mole}$	2024
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2. xii) Calculate standard enthalpy of formation ΔH_f° of $(C_3H_8O_3)$ from the data given below. * $3C + 4H_2 + \frac{3}{2}O_2 \rightarrow C_3H_8O_3$ $\Delta H_f^\circ = ?$ * $C + O_2 \rightarrow CO_2$ $\Delta H_f^\circ = -393.5 \text{ KJ/mole}$ * $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ $\Delta H_f^\circ = -285.8 \text{ KJ/mole}$ * $C_3H_8O_3 + 3\frac{1}{2}O_2 \rightarrow 3CO_2 + 4H_2O$ $\Delta H_f^\circ = -1654.1 \text{ KJ/mole}$	2023
(vii) Calculate the heat of formation of benzene at 25°C: * $6C + 3H_2 \longrightarrow C_6H_6$ $\Delta H_f = ?$ * $C + O_2 \longrightarrow CO_2$ $\Delta H_f = -394 \text{ kJ/mol}$ * $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$ $\Delta H_f = -286 \text{ kJ/mol}$ * $C_6H_6 + \frac{15}{2}O_2 \longrightarrow 6CO_2 + 3H_2O$ $\Delta H_f = -3267 \text{ kJ/mol}$	2022
2. v) Calculate ΔH_f for the following data: * $6C + 3H_2 \longrightarrow C_6H_6$ $\Delta H_f = ?$ * $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$ $\Delta H_f = -286 \text{ kJ/mol}$ * $C + O_2 \longrightarrow CO_2$ $\Delta H_f = -394 \text{ kJ/mol}$ * $C_6H_6 + \frac{15}{2}O_2 \longrightarrow 6CO_2 + 3H_2O$ $\Delta H_f = -3267 \text{ kJ/mol}$ 2. ix) State first law of thermodynamics and show that: * $q_p = \Delta H$ * $W = P\Delta V$	2021
(ix) Calculate the Heat of formation of Fe_2O_3 at 25°C. * $2Fe + \frac{3}{2}O_2 \longrightarrow Fe_2O_3$ $\Delta H_f = ?$ * $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$ $\Delta H = -256 \text{ KJ/mole}$ * $Fe_2O_3 + 3H_2O \longrightarrow 2Fe(OH)_3$ $\Delta H = +289 \text{ J/mole}$ * $Fe + 3H_2O \longrightarrow Fe(OH)_3 + \frac{3}{2}H_2$ $\Delta H = +161 \text{ J/mole}$ (x) State and explain Hess's Law of Constant heat summation with its applications.	2019
(ix) State the First Law of thermodynamics and show that: $q_v = \Delta E$ $q_p = \Delta H$ (x) Calculate $\Delta H_{25^\circ\text{C}}$ of ethane from the data given below * $2C_{(s)} + 3H_{2(g)} \longrightarrow C_2H_{6(g)}$ $\Delta H_{25^\circ\text{C}} = ?$ * $C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$ $\Delta H_{25^\circ\text{C}} = -394 \text{ KJ/mole}$	2018

* $H_{2(g)} + \frac{1}{2}O_{2(g)} \longrightarrow H_2O(l) \quad \Delta H_{25^\circ C} = -286 \text{ KJ / mole}$ * $C_2H_{4(g)} + \frac{7}{2}O_{2(g)} \longrightarrow 2CO_{2(g)} + 3H_2O(l) \quad \Delta H_{25^\circ C} = -1560 \text{ KJ / mole}$	
(viii) Calculate the heat of formation of Benzene at $25^\circ C$. * $6C + 3H_{2(g)} \longrightarrow C_6H_{6(l)} \quad \Delta H_f = ?$ * $C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)} \quad \Delta H_f = -394 \text{ KJ}$ * $H_{2(g)} + \frac{1}{2}O_{2(g)} \longrightarrow H_2O(l) \quad \Delta H_f = -286 \text{ KJ}$ * $C_4H_{6(l)} + \frac{15}{2}O_{2(g)} \longrightarrow 6CO_{2(g)} + 3H_2O(l) \quad \Delta H_f = -3267 \text{ K}$ (ix) State and explain Hess's Law of constant heat of summation with examples.	2017
(v) Calculate the heat of formation of M_2A AT $25^\circ C$: $2M + \frac{1}{2}A_2 \longrightarrow M_2A \quad (\Delta H = ?)$ $M + Y_2A \longrightarrow MAY + \frac{1}{2}Y_2 \quad (\Delta H = -43.2 \text{ KJ/Mole})$ $Y_2 + \frac{1}{2}A_2 \longrightarrow Y_2A \quad (\Delta H = -58.5 \text{ KJ/Mole})$ $M_2A + YA_2 \longrightarrow 2MAY \quad (\Delta H = -63.2 \text{ KJ/Mole})$ (ix) (b) If 1800 cal of heat is added to a system while the system does work equivalent to 2800 cal by expanding against the surrounding, what is the value of ΔE for the system?	2016
(v) Calculate the heat of formation of CH_3OH from the following data: * $C + 2H_2 + \frac{1}{2}O_2 \longrightarrow CH_3OH \quad \Delta H_f = ?$ * $C + \frac{1}{2}O_2 \longrightarrow CO \quad \Delta H_f = -111 \text{ KJ/mole}$ * $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O \quad \Delta H_f = -286 \text{ KJ/mole}$ * $CH_3OH + O_2 \longrightarrow CO + 2H_2O \quad \Delta H_f = -561 \text{ KJ/mole}$ (xv) What 400 J of heat is added to a gaseous system at a constant pressure of 101300 N/m^2, its Internal energy increases by 500 J. Calculate the change in the volume of the system.	2015
(vii) Calculate the heat of formation from the following data: * $4AX_3 + 5Y_2 \longrightarrow 4AY + 6X_2Y \quad \Delta H_f = ?$ * $\frac{1}{2}A_2 + \frac{3}{2}X_2 \longrightarrow AX_3 \quad \Delta H = -11.0 \text{ KJ/mole}$ * $X_2 + \frac{1}{2}Y_2 \longrightarrow X_2Y \quad \Delta H = -57.8 \text{ KJ/mole}$ * $\frac{1}{2}A_2 + \frac{1}{2}Y_2 \longrightarrow AY \quad \Delta H = +21.6 \text{ KJ/mole}$	2014
Section C	

4. a) State Hess's Law? What is meant by Born-Haber's Cycle? Determine the lattice energy for CsF , writing all the involved thermochemical equations stepwise, using following data: (all values are in KJ/mole) ΔH_f° of $\text{CsF} = -553.5$ ΔH_{sub} of $\text{Cs} = 76.5$ $\Delta H_{B.A}$ of $\text{F} = -328.2$ ΔH_{sub} of $\text{Cs} = 375.5$ ΔH_{diss} of $\text{F}_2 = 157$	2023
4. b) Give the statement and mathematical derivation of First Law of Thermodynamics and derive $q_v = \Delta E$ and $q_p = \Delta H$	
5.(a) State and explain the First Law of Thermodynamics. Show that $W = P\Delta V$ and $\Delta H = \Delta E + P\Delta V$	2016
3(c) Define Thermochemistry. State and explain Hess's law of Constant heat summation with its applications.	2015
4. (b) State and explain the First Law of Thermodynamics also prove the $q_p = \Delta E + P\Delta V = \Delta H$	2014

Chapter 12: ELECTROCHEMISTRY

Section B

(xiii) What is Electrolysis? Explain electrolysis in CuCl_2 solution. Also give the chemical reactions on the electrodes.	2018
4. (c) Balance any one of the following equations by ion Electron methods. $\text{I}^- + \text{Cr}_2\text{O}_7^{2-} + \text{H}^+ \rightarrow \text{Cr}^{3+} + \text{IO}_3^- + \text{H}_2\text{O}$ (Acidic Medium) $\text{Cr}(\text{OH})_3 + \text{SO}_4^{2-} \rightarrow \text{CrO}_4^{2-} + \text{SO}_3^{2-}$ (Basic Medium)	2017
(xiii) Enumerate the rules of assigning oxidation number of the elements.	2016
(x) Balance the following equation by Ion-electron method: $\text{Fe}^{+3} + \text{Cr}_2\text{O}_7^{2-} + \text{H}^+ \rightarrow \text{Fe}^{+2} + \text{Cr}^{+3} + \text{H}_2\text{O}$	2015
(ix) Balance the following equations by Ion-electron method: (i) $\text{Cl}_2 + \text{OH}^- \rightarrow \text{ClO}_3^- + \text{H}_2\text{O}$ (ii) $\text{HNO}_3 + \text{H}_2\text{S} \rightarrow \text{NO} + \text{S} + \text{H}_2\text{O}$	2014

Section C

3. b) Define Corrosion. Balance the following equations by ion-electron method: $\text{Cr}(\text{OH})_3 + \text{SO}_4^{2-} \rightarrow \text{CrO}_4^{2-} + \text{SO}_3^{2-}$ (Basic medium) $\text{Cr}_2\text{O}_7^{2-} + \text{I}_2 \rightarrow \text{Cr}^{3+} + \text{IO}_3^-$ (Acidic medium)	2024
5. a) Define battery. Explain the working of lead storage battery and determine electrode potential of copper or zinc.	

5. a) Define Oxidation and Reduction according to modern electronic concept. Balance the following equation by ion electron method: * $Cr_2O_7^{2-} + Cl^- \rightarrow + Cr^{+3} + Cl_2$ (Acidic medium) * $MnO_4^- + SO_3^{2-} \rightarrow Mn^{+2} + SO_4^{2-}$ (Basic medium) 5. b) Define Cell-Potential, Standard Electrode Potential. Explain the determination of standard electrode Zinc with diagram and potential of cell reactions.	2023
4. (a) Define standard electrode potential. What is standard hydrogen electrode and How is the electrode potential of Zn determined? 4. (b) Balance any two of the following equations by ion electron method: * $Cr_2O_7^{2-} + I_2 \longrightarrow Cr^{3+} + IO_3^{-1}$ Acidic medium * $MnO_4^- + SO_3^{2-} \longrightarrow Mn^{2+} + SO_4^{2-}$ Acidic medium * $Br_2 + OH^- \longrightarrow Br^- + BrO_3^- + H_2O$ Basic medium	2022
3. b) Balance the following equations by ion electron method: * $Cr_2O_7^{2-} + I_2 \longrightarrow Cr^{3+} + IO_3^{-1}$ Acidic medium * $MnO_4^{-1} + SO_3^{2-} \longrightarrow Mn^{+2} + SO_4^{2-}$ Basic medium 5. a) Write the postulates of Arrhenius Theory of Ionization calculate Oxidation Number of the following: * Cr in $Cr_2O_7^{2-}$ * S in $H_2S_2O_7$ * N in NH_4^+	2021
5. (a) What is Standard Electrode potential? How is the electrode potential of Copper determined? Explain 5. (b) Balance any One of the following by ion-electron method: * $Cr(OH)_3 + SO_4^{2-} \longrightarrow CrO_4^{2-} + SO_3^{2-}$ (Basic Medium) * $MnO_4^- + Cl^- \longrightarrow Mn^{2+} + Cl_2$ (Acidic Medium)	2019
5. (a) What is Electrode potential? Determine the electrode potential of Zinc with the help of cell diagram 5. (b) Balance any one of these equation by ion electron method: * $HNO_3 + H_2S \longrightarrow NO + S + H_2O$ * $MnO_4^- + SO_3^{2-} + OH^- \longrightarrow Mn^{+2} + SO_4^{2-}$	2018
4. (a) What is Electrolysis? Explain by giving the example of $CuCl_2$. Give all the necessary electrode reactions.	2017
5 (b) Define standard Electrode potential. How is the electrode-potential of Zn determined?	2016

3(b) What is standard electrode potential? How is the electrode potential of Copper determined? Explain.	2015
3. (b) What is meant by Electrolysis? Explain the electrolysis of CuCl_2 solution giving all the necessary electrode reactions.	2014
5. (b) Define Electrode potential. What is standard electrode potential? How is the electrode potential of Copper determined?	

MISCELLANEOUS QUESTIONS

DEFINE / DEFINITIONS

QUESTIONS	CHAPTER	YEAR
2. i) Define any four of the following: *Significant Figure	1	2025
*Rounding Off Data	1	
*Bond Energy *Dipole Moment	3	
2. i) Define any four of the following: * Stoichiometry * Exponential notation	1	2024
* Unit Cell	6	
* Latent Heat of Fusion	5	
* Energy of Activation	9	
2. i) Define any four of the following: * Molar Volume Limiting Reactant	1	2023
* Bond Energy * Dipole Moment	3	
* Evaporation	5	
2. i) Define the following: *Avogadro's Number *Limiting Reactant	1	2022
*Heat of Formation	11	
*Rate Expression	9	
2. i) Define the following: * Significant Figure * Molar Volume	1	2021
* Viscosity	5	
* Catalyst	9	
2. (i) Define the following:		2019
(i)Random error (ii)Exponential notation	1	
(iii)Heat of formation	11	
(iv) Rate expression	9	
2. (i) Define the following:		2018
*Stoichiometry *Molar volume	1	
*Activation energy	9	
*Enthalpy	11	